

Reply to: “Tailoring allocation policies and improving access to pediatric liver transplantation in Italy: Outframing and concluding”

To the Editor:

We thank Jean de Ville de Goyet¹ for his interest in our work recently published in the *Journal of Hepatology*.² The objective of our study was to define the outcome of patients <18 years old listed in Italy for liver transplantation (LT) from deceased donors in an intention-to-treat (ITT) survival analysis and to identify the factors that influence the possibility of being transplanted and leaving the waiting list (WL) due to death or clinical deterioration. The results of our study showed that children on the Italian deceased-donor LT (DDLT) WL are transplanted within 44 days (IQR 15–120) in case of status 3 (non-urgent transplantation) and within 2 days (IQR 1–5) for status 1 (urgent transplantation). The continuous adaption of liver allocation policies for pediatric LT led to satisfactory outcomes, with ITT survival rates of 90.5% at 1 year and 87.5% at 5 years, remaining stable across Eras.

In our analysis, children who were listed for a DDLT but eventually received a living-donor LT (LDLT) were included in the description of the pediatric WL dynamics, but they were censored for the outcome analysis as the investigation of LDLT was not the objective of the current study. In the analysis of the organ allocation process, these LT candidates had to be considered, as well as patients dropped from the LT WL for clinical improvement, because they actively competed for the assignment of an organ until they were removed from the WL. Clearly, the fact that they were transplanted using a living-related liver graft increased the availability of deceased-related organs for the other LT candidates, but this is only a consideration which can be made in hindsight. Hence, the conclusion and statistics presented in De Ville de Goyet's letter are not evidence based: the numbers are not real, and an assumption shouldn't be based on an approximate estimation.

In our paper, we also strongly stated that living-related LT is a complementary resource to DDLT for pediatric candidates and that it should be encouraged even in effective deceased-donor allocation systems like the Italian transplant organization. In our country, during the study period (2002–2018), the proportion of LDLT increased from 0.9% in Era 1 to 11.8% in Era 3, and up to 23.3% in the last observed year (2018). Thus, contrary to the expectation, after the initial increase in the proportion of LDLTs between 2019–2022, the proportion (75 LDLTs/344 total LTs) didn't increase further, being steady at 21.8% of total pediatric LTs performed every year, while in the same period 52% (179 split LT/344 total LT) of children were transplanted with deceased-donor split grafts.

As the author of the letter is aware, in Italy, children who are not resident in Italy and not registered in the Italian healthcare

system cannot receive a primary LT from a deceased donor, except in the presence of specific agreements between states. These patients can receive a transplant from a living donor and can only benefit from a graft from a deceased donor in case of urgent need for retransplantation. For this reason, patients are placed on the WL a few days before the living-donor transplant is scheduled. Eleven of the 92 (12%) LDLTs carried out during the period analyzed in our study were performed in children not registered within the Italian healthcare system; therefore, 81 (6.3%) patients receiving LDLT influenced the DDLT activity. Indeed, in Era 3, 36 (13.1%) patients received LDLT. Between 2019 and 2022, of 75 LDLTs, 18 (16.6%) were performed in children not registered with the national health system.

For all the above reasons, the author cannot conclude that the satisfactory outcomes observed for the pediatric LT WL are a direct consequence of increased LDLT activity in Italy, as this subgroup of patients would not use the Italian graft resource. Rather, we believe that because of the relatively small number of pediatric LTs performed each year, fundamentally based on pediatric liver demand, LT dynamics are subject to rapid changes in accordance with the policies adopted by a single transplant center. Further study and ongoing analysis of the Italian experience in pediatric LDLT will better clarify the role of living donation in LT.

In our study, we also identified the categories of LT candidates at higher risk of not receiving LT (including recipient's weight >25 kg, blood group O or awaiting LT combined with other organs) and mortality in an ITT analysis (re-transplantation (HR 5.83, $p < 0.001$), Status 1 (HR 2.28, $p = 0.006$), Status 1B (HR 2.90, $p = 0.014$), Status 2A (HR 9.12, $p < 0.001$), recipient weight <6 kg (HR 4.53, $p < 0.001$) and low-volume activity (HR 4.38, $p = 0.001$). Among the combination of risk factors, we observed that WL mortality occurred mainly in very sick patients listed as status 3 with a high PELD/MELD score, such as children with acute-on-chronic liver failure or requiring re-transplantation. Therefore, as discussed in our paper and known to the entire Italian transplant community, since the 25th June 2020, the College of Italian Pediatric Transplant Programs in collaboration with the National Transplant Center have introduced several modifications³ of the pediatric liver allocation policies in order to improve outcomes of such frail children and reduce the number of “authorized urgency”:

- Pediatric LT candidates with PELD/MELD scores ≥ 35 are prioritized to status 2A (instead of status 3) in the pediatric national WL, as in the macro-area adult allocation. In the macro-area allocation, children receive organ offers from a) adult donor potentially splittable (≤ 50 years at standard risk) not only as

left-lateral split graft but also as extended-right graft or whole liver graft, and b) from donors older than 50 years and/or at non-standard risk (which they would not otherwise receive).

- Status-2A patients receive graft offers also from donors with compatible and (if needed) incompatible blood groups, as occurs in status-1 candidates, because status-2A is considered urgent transplantation. Hence, we believe that this category should be considered as status-1 patients in terms of allocation priorities, as already proposed in our paper.
- Pediatric candidates for late (>1 year) re-LT with PELD/MELD scores >20 are prioritized to status 2B.

These corrections also aimed to give priority to more pediatric candidates over those with stable metabolic diseases; moreover, rules and strategies to promote the full-left/full-right split in favor of recipients weighing more than 25 kg are under discussion. The effectiveness of these measures is currently being analyzed. Therefore, most of the suggested improvements have already been implemented thanks to the data analysis and the constant adaptation process, which continuously involves all Italian pediatric transplant programs, including the author of the letter. Nonetheless, as part of the ongoing review of allocative policies promoted by the National Transplant Center, the results of our study, as well as those of other recent Italian multicenter studies,⁴ will be used to introduce further improvements to the Italian pediatric transplant program.

In conclusion, we believe that analyses of LT WL dynamics, like the one performed in our study, are performed with the sole intention of improving outcomes for disadvantaged pediatric LT candidates. We thank the author for his comments which confirmed our conclusions, bringing to light that, as in the past and also in the future, the Italian pediatric transplant community must work together to minimize to almost zero pediatric WL mortality. The Italian experience suggests that the pediatric liver graft allocation is a dynamic process influenced not only by donor and recipient features, but also by centers' choices, and it requires continuous adaption to optimize outcomes of pediatric LT candidates.

Marco Spada^{1,*}
 Roberta Angelico²
 Silvia Trapani³
 Lucia Masiero³
 Francesca Puoti³
 Michele Colledan^{4,5}
 Davide Cintonio⁶
 Renato Romagnoli⁷
 Umberto Cillo⁸
 Massimo Cardillo³

¹Division of Hepatobiliopancreatic Surgery, Liver and Kidney Transplantation, Research Unit of Clinical Hepatogastroenterology and Transplantation, Bambino Gesù Children's Hospital, IRCCS, Rome, Italy

²HPB and Transplant Unit, Department of Surgical Sciences, University of Rome Tor Vergata, Rome, Italy

³Italian National Transplant Center, National Institute of Health, Rome, Italy

⁴Department of Organ Failure and Transplantation - ASST Papa Giovanni XXIII, Bergamo, Italy

⁵Università Milano-Bicocca, Milan, Italy

⁶Department of Pediatrics for the Study of Abdominal Diseases and Abdominal Transplantation, ISMETT (Istituto Mediterraneo per i Trapianti e Terapie ad Alta Specializzazione), IRCCS -UPMC (University of Pittsburgh Medical Center), Palermo, Italy

⁷General Surgery 2U, Liver Transplant Unit, A.O.U. Città della Salute e della Scienza di Torino, University of Torino, Torino, Italy

⁸Hepatobiliary Surgery and Liver Transplantation Unit, University of Padova, Padova, Italy

*Corresponding author. Address: Division of Abdominal Transplantation and Hepatobiliopancreatic Surgery, Bambino Gesù, Children's Research Hospital IRCCS, Piazza Sant'Onofrio 4, 00165 Rome, Italy; Tel.: 0039 06 68591, fax: 0039 06 68591. E-mail address: marco.spada@opbg.net (M. Spada)

Received 1 May 2024; Accepted 7 May 2024; Available online 15 May 2024

<https://doi.org/10.1016/j.jhep.2024.05.011>

© 2024 European Association for the Study of the Liver. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



Financial support

The authors did not receive any financial support to produce this manuscript.

Conflict of interest

The authors of this study declare that they do not have any conflict of interest. Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Conceptualization, M.S., R.A., S.T.; methodology, M.S., R.A., L.M., F.P.; validation, L.M., F.P., S.T.; formal analysis, L.M., F.P.; investigation, R.A., S.T.; data curation, L.M., F.P., S.T.; writing—original draft preparation, M.S., R.A.; writing—review and editing, M.C., D.C., R.R., U.C.; supervision, M.C.. All authors have read and agreed to the published version of the manuscript.

Data availability statement

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.05.011>.

References

Author names in bold designate shared co-first authorship

- [1] de Ville de Goyet J, Spada M, et al. Tailoring allocation policies and improving access to pediatric liver transplantation in Italy: outframing and concluding: comment on: tailoring allocation policies and improving access to pediatric liver transplantation over a 16-year period. *J Hepatol* 2024 Feb 12;81:e120–e121. S0168-8278(24)00126-0.
- [2] **Spada M, Angelico R, Trapani S**, et al. Tailoring allocation policies and improving access to paediatric liver transplantation over a 16-year period. *J Hepatol* 2024 Mar;80(3):505–514.
- [3] <https://www.trapianti.salute.gov.it/trapianti/archivioProtocolliCnt.jsp?lingua=italiano&anno=2021&anno=2020&btnCerca=cerca>. Accessed on April 30th, 2024.
- [4] Lauterio A, Cillo U, Spada M, et al. Improving outcomes of in situ split liver transplantation in Italy over the last 25 years. *J Hepatol* 2023 Dec;79(6):1459–1468.